

Civics Group	Index Number	Name (use BLOCK LETTERS)

H2



**ST. ANDREW'S JUNIOR COLLEGE
2014 JC2 Preliminary Examinations**

H2 BIOLOGY

9648/1

Paper 1: Multiple Choice

Thursday

18 September 2014

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser (not supplied)
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the multiple choice answer sheet in the spaces provided.

There are **40** questions in this paper. Answer all questions. For each question, there are four possible answers, A, B, C and D.

Choose the one you consider correct and record your choice in soft pencil on the separate multiple choice answer sheet.

INFORMATION TO CANDIDATES

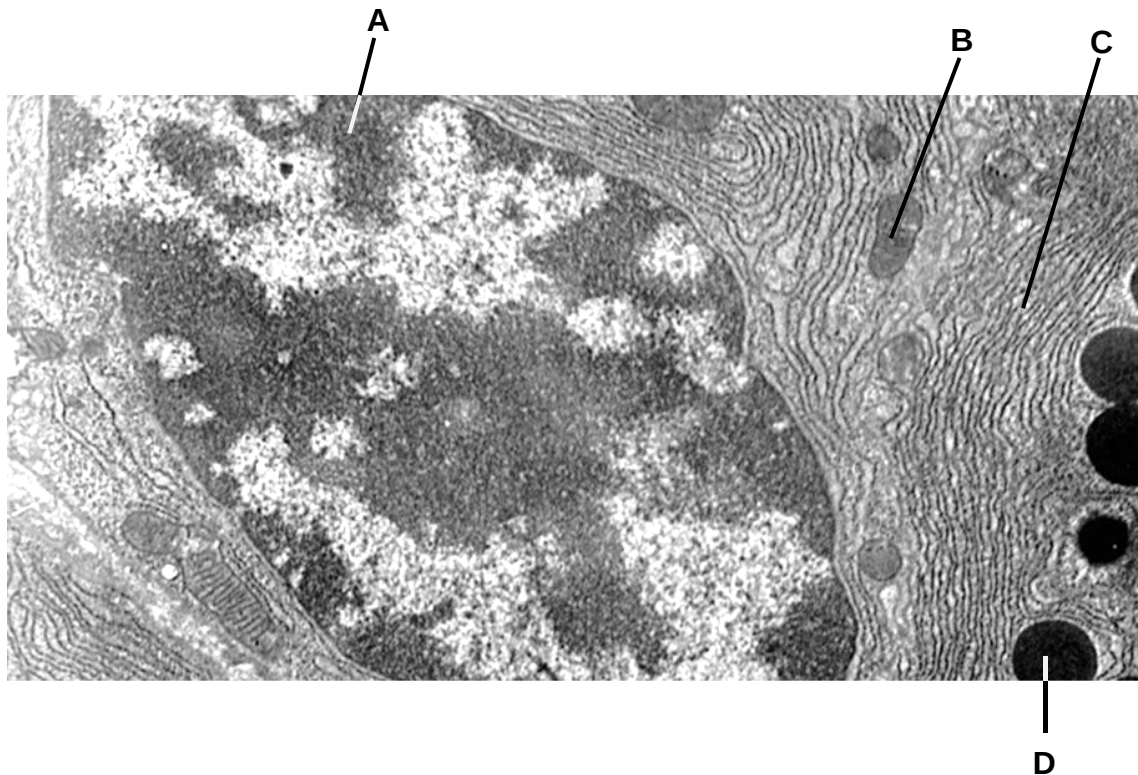
Each correct answer will score one mark. A mark will not be deducted for wrong answer. Any rough working should be done in this booklet.

At the end of the examination, submit the multiple choice answer sheet only.

This document consists of **23** printed pages.

[Turn over

- 1 The electron micrograph below shows a beta cell from the islets of Langerhans in the pancreas.

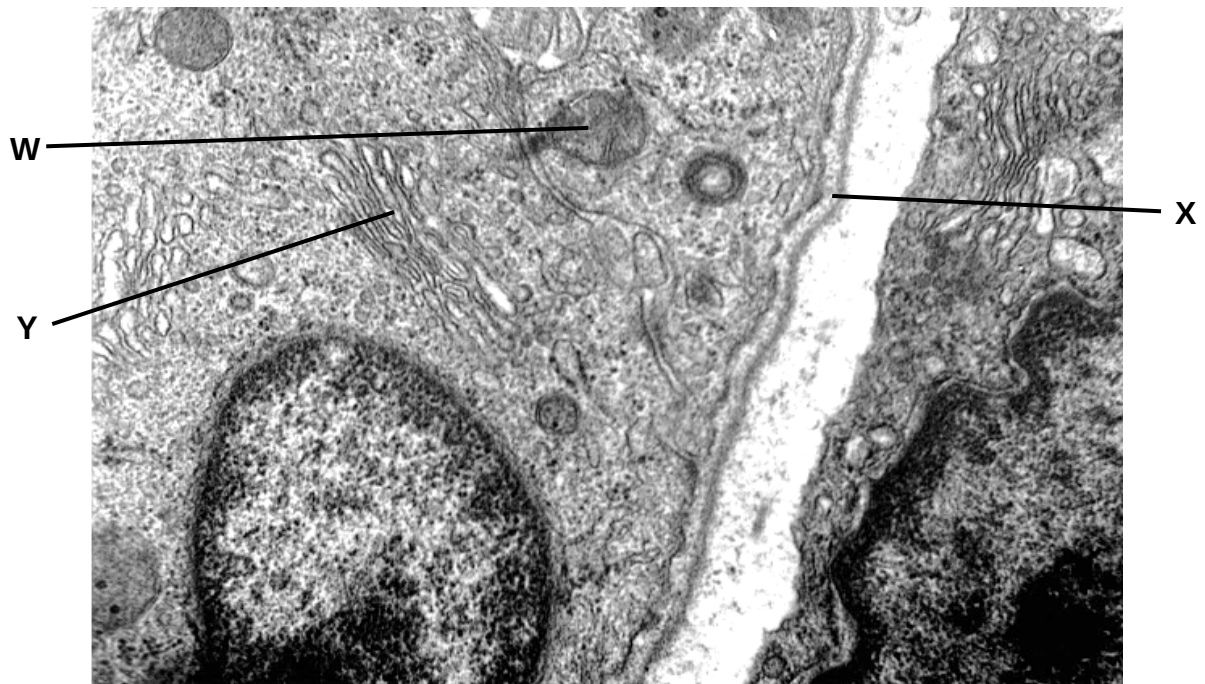


Which statements correctly describe the labelled structures?

- 1 Structure **A** has transcriptionally active regions which are accessible to enzymes for transcription of insulin mRNA.
- 2 Structure **B** is the site of cellular respiration, providing ATP necessary for the synthesis of glucagon.
- 3 Proteins enter the lumen of Structure **C**, where they undergo chemical modifications such as glycosylation.
- 4 When Benedict's reagent is added to Structure **D**, the Benedict's reagent would form a brick red precipitate.

- A** 1 only
B 3 only
C 1 and 3
D 1, 2 and 4

- 2 The transmission electron micrograph shows part of two cells.

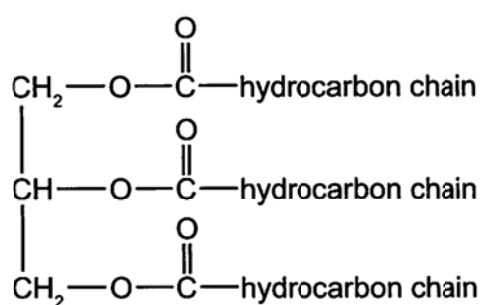


Which of the labeled features enable these cells to be identified as eukaryotic?

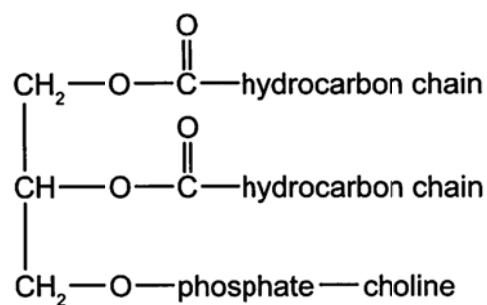
- A W only
 - B W and X
 - C W and Y
 - D W, X and Y
- 3 Which of the statements about polysaccharides can be used to describe both amylopectin and cellulose.
- 1 adjacent glucose molecules are rotated by 180°
 - 2 contains 1,4 glycosidic bonds
 - 3 polymer of α -glucose

- A 2 only
- B 3 only
- C 1 and 2
- D 1 and 3

- 4 The diagram shows the structure of two lipid molecules.



X



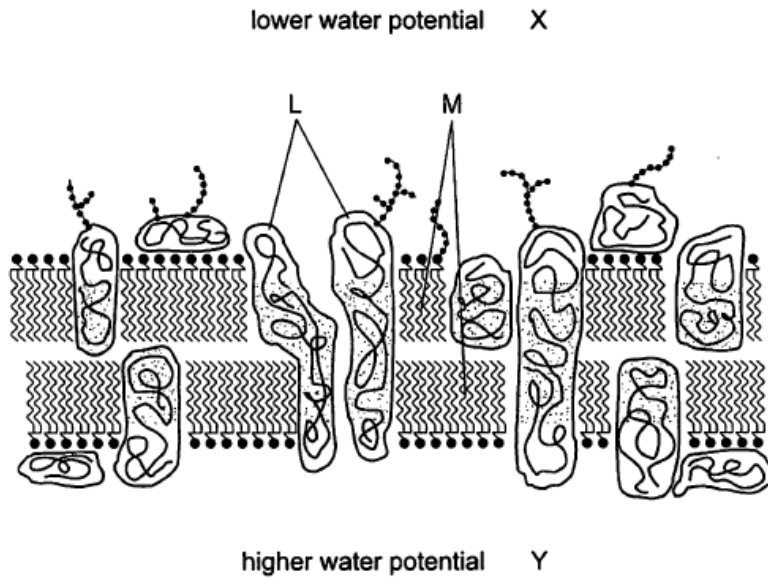
Y

Which statements correctly describe the two lipid molecules?

- 1 Both molecules are esters of glycerol.
- 2 Both the molecules are made by condensation reactions.
- 3 The hydrocarbon chains of molecule X are always from saturated fatty acids.
- 4 The hydrocarbon chains of molecule Y may be from saturated or unsaturated fatty acids.
- 5 The hydrocarbon chains of molecule Y always the same length.

- A** 1 only 2
B 1 only 3
C 1, 2 and 4
D 2, 3 and 5

- 5 The diagram shows part of the plasma membrane of an animal cell.

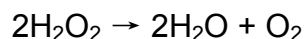


Which statements correctly describe the movements of molecules across this membrane?

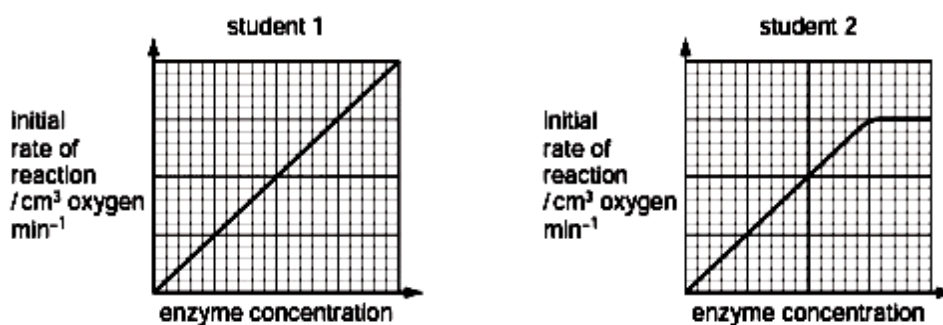
- 1 Oxygen diffuses through molecules M from X to Y.
- 2 Carbon dioxide diffuses through molecules M from Y to X.
- 3 Molecule L has a hydrophobic water channel for movement of water by osmosis.
- 4 There is a net movement of water from Y to X through the hydrophilic channel of molecule L.

- A** 4 only
B 1 and 3
C 1, 2 and 4
D 1, 2, 3 and 4

- 6 Catalase is an enzyme that catalyses the conversion of hydrogen peroxide into water and oxygen.



Two students investigated the effect of enzyme concentration on the rate of reaction of the enzyme catalase. The students predicted their results would show the same trend. The graphs show the rates obtained by each student.



Which statement explains the different trend shown by student 2's results?

- A Student 2 included a competitive inhibitor in the investigation.
 - B Student 2 performed the investigation at a higher temperature.
 - C Student 2 performed the investigation at pH 6 compared to pH 8.
 - D Student 2 used a lower concentration of substrate in the investigation.
- 7 The diagram shows the chromosomes of one cell during mitosis.



Which stage of mitosis is shown and what is the haploid number in this species?

	Stage of mitosis	Haploid number
A	Anaphase	5
B	Anaphase	10
C	Metaphase	5
D	Metaphase	10

- 8 A short piece of DNA 19 base pairs long was analysed to find the number of nucleotide bases in each of the polynucleotide strands. Some of the results are shown below.

	number of nucleotide bases			
	A	C	G	T
strand 1				4
strand 2		7		5

How many nucleotides containing C were present in strand 1?

- A** 2
B 3
C 5
D 7
- 9 The mechanism of action of four drugs that inhibit DNA replication is stated below.

- Aphidicholine inhibits DNA polymerase.
- Cytarabine is converted into a molecule that can substitute for a DNA nucleotide and also inhibits DNA repair mechanisms.
- Epirubicin inhibits an enzyme involved in the unwinding of DNA and separation of strands.
- Hydroxycarbamide inhibits an enzyme involved in the production of deoxyribonucleotides.

Which row correctly matches a drug to an explanation of the mechanism of action?

	Explanation of mechanism of action			
	Decreased pool of available nucleotides inhibits chain elongation	DNA strands not available as templates for replication	DNA damaged during replication and cell death occurs	Exposed DNA template strands unable to be copied
A	Aphidicholine	Epirubicin	Cytarabine	Hydroxycarbamide
B	Epirubicin	Cytarabine	Hydroxycarbamide	Aphidicholine
C	Hydroxycarbamide	Aphidicholine	Epirubicin	Cytarabine
D	Hydroxycarbamide	Epirubicin	Cytarabine	Aphidicholine

- 10** One individual, organism P, in a sexually-reproducing population of mammals has two forms of a protein in its heart muscle. The normal form is identical to that in most other members of the population, while the variant form differs by one amino acid. The diagram shows the amino acid sequences in part of the normal protein and in the same part of the variant protein.

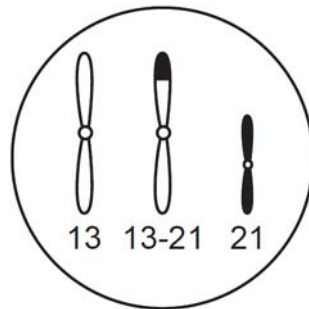
normal protein	variant protein
-glu-lys-lys-gly-glu-	-glu-lys-gly-gly-glu-

- 1 The sequence of amino acids was altered in a cell in the heart muscle of P's mother.
- 2 The sequence of amino acids was altered in a cell in the ovaries of P's mother.
- 3 The sequence of bases was altered in the DNA in a cell in the heart muscle of P.
- 4 The sequence of bases was altered in the DNA in a cell in an ovary of P.
- 5 The sequence of bases was altered in the DNA in a cell in a testis of P's grandfather.

What could account for the presence of the normal protein and the variant protein in P?

- A** 1, 2 and 4
- B** 1 and 3
- C** 2, 4 and 5
- D** 3 and 5

- 11 Down's syndrome can be caused by a trisomy of chromosome 21, but can also result from translocation of chromosome 21 onto chromosome 13, forming a single chromosome 13-21.



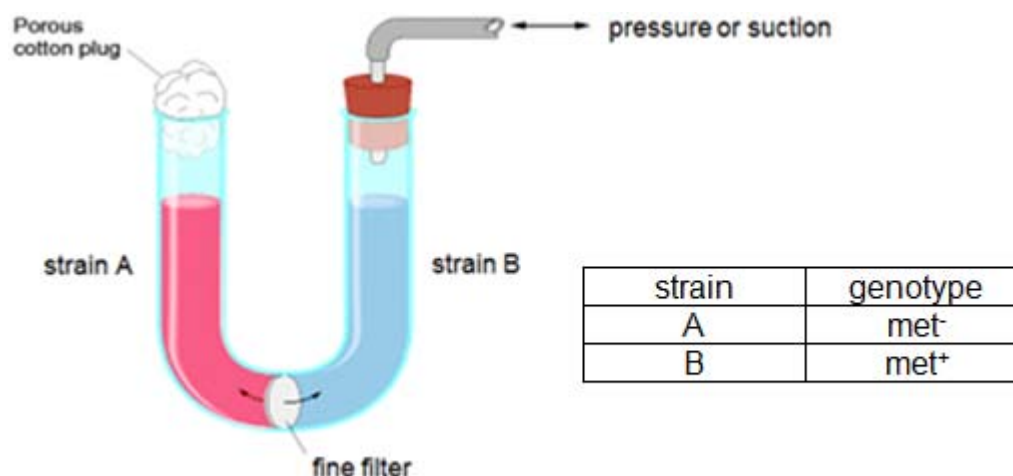
The diagram shows chromosomes 13 and 21 in the nucleus of a diploid ($2n$) testis cell from a phenotypically normal male carrier of a 13-21 translocation. This cell has a chromosome number of 45.

Which is **not** a likely outcome of fertilisation of normal oocytes by sperm from this male?

	Chromosomes in sperm	Embryo
A	13 and 21	$2n = 46$ normal phenotype
B	13-21	$2n = 45$ normal phenotype
C	13-21 and 21	$2n = 46$ Down's syndrome
D	13-21 and 21	$2n = 47$ Down's syndrome

- 12 With reference to prokaryotic and eukaryotic genomes, which of the following statement is **not** true?
- A** They both have extra-chromosomal DNA.
 - B** They both have non-coding regions.
 - C** They both have regulatory sequences.
 - D** They are both associated with histones.

- 13 To investigate gene transfer between bacteria, two strains of the same bacterial species were each placed in one arm of a U-tube with a filter separating them.



met⁺ is a wild-type gene that codes for the ability to synthesise the essential amino acid, methionine.

met⁻ indicates that the met⁺ gene has been mutated.

Liquid may be transferred between the arms of the tube by the application of pressure or suction, but particles that are larger than the filter pore size would not be able to pass through the fine filter.

type of particle	size
bacteria	1 – 10µm
bacteriophages	0.025 – 0.2µm

After several hours of incubation, bacterial cells from the left arm of the tube are plated on minimal medium.

Which pair of experimental results best shows that transduction was most likely the process responsible for gene transfer between strains A and B?

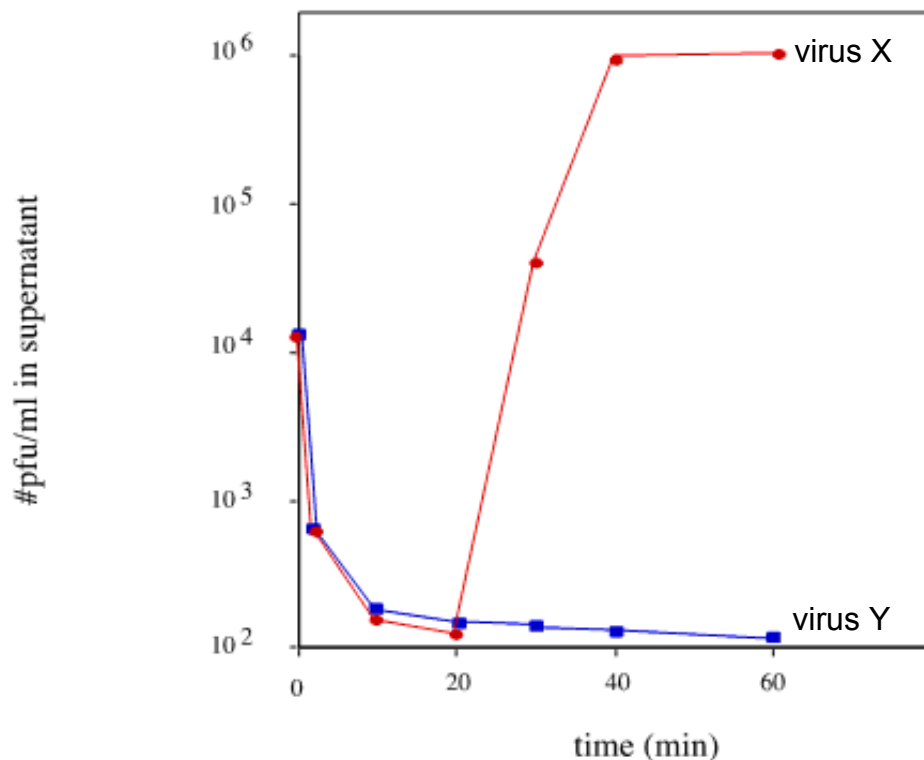
	filter pore size	growth of colonies on minimal medium
A	5µm	no
	0.1µm	yes
B	5µm	no
	0.1µm	no
C	0.45µm	no
	0.02µm	yes
D	0.45µm	yes
	0.02µm	no

14 How do viruses cause diseases in animals?

- 1 They degrade the host cell's chromosomes.
- 2 They disrupt the oncogenes of the host cell causing uncontrolled cell division.
- 3 Their viral proteins and glycoproteins on the surface membrane of host cells cause them to be recognized and destroyed by the body's immune defenses.

- A 1 and 2
 B 1 and 3
 C 2 and 3
 D 1, 2 and 3

15 Two types of viruses, X and Y, were added to a culture of bacteria. For each type of virus, the change in the number of infectious virus particles present in the supernatant (pfu/ml in supernatant) was monitored for 60 minutes and shown in the graph.

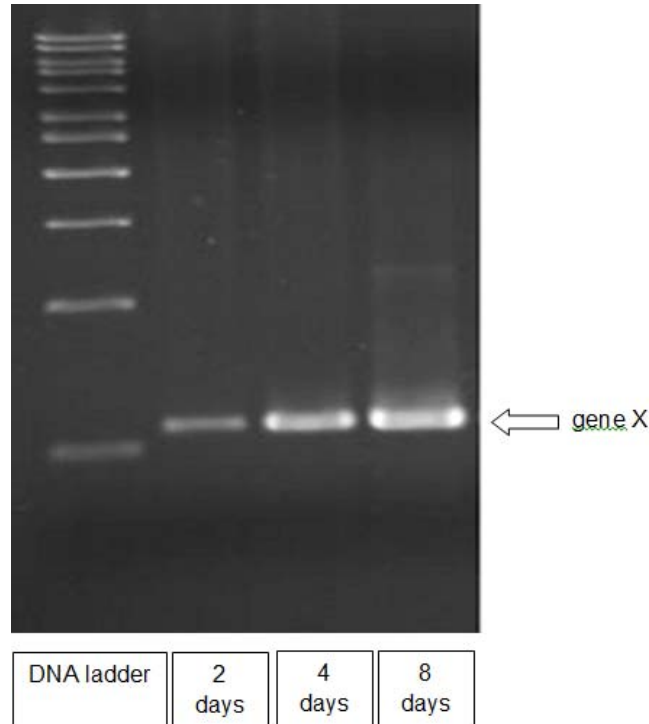


What could explain the sharp increase in the number of infectious virus X particles present in the supernatant from 20 to 40 minutes?

- A injection of viral DNA into host cell
 B integration of viral DNA into host cell DNA
 C release of viral particles by cell lysis
 D release of viral particles by budding

- 16** Gel electrophoresis was performed using samples of gene X isolated from equal number of cells from a human embryo after 2 days, 4 days and 8 days of development.

What kind of gene regulation is illustrated by the results of gel electrophoresis?



- A** DNA methylation
B gene amplification
C histone acetylation
D transcriptional activation
- 17** In a laboratory, the 5' 7-methylguanosine cap and the length of the polyA tail of an mRNA transcript are altered in order to examine their effects on the number of functional proteins translated.

mRNA	Alteration
1	5' cap removed with 1000-adenine tail
2	5' cap removed with 500-adenine tail
3	5' cap present with 500-adenine tail
4	5' cap present with 300-adenine tail

Using the information given in the table above, predict the relative amount of protein produced in descending order (greatest → least).

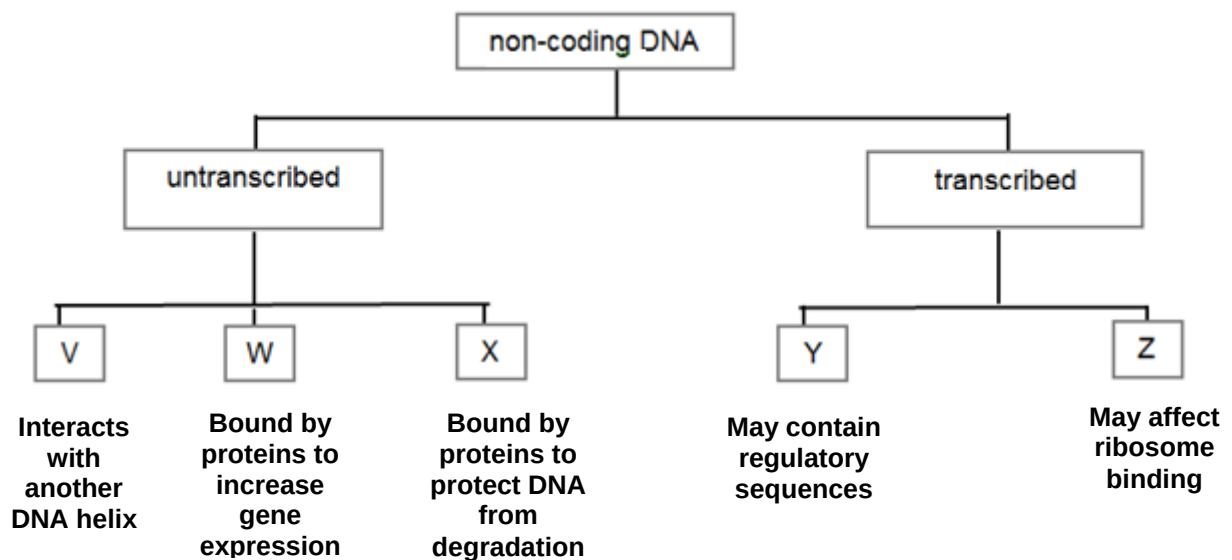
- A** 1,3,2,4
B 1,2,3,4
C 3,1,4,2
D 3,4,1,2

18 Which of the following statement about chromosome structure is true?

- 1 Linker DNA refers to the linear double-stranded DNA between adjacent nucleosomes.
- 2 The 30 nm chromatin fibre has a solenoid structure.
- 3 The 300 nm chromatin fibre is attached to multiple locations on a central protein scaffold.
- 4 Euchromatin is the more diffuse region of the interphase chromosome and is transcriptionally inactive.

- A** 1 and 2
B 1 and 4
C 2 and 3
D 3 and 4

19 The diagram shows the classification of several portions of eukaryotic DNA that do not encode RNA or protein.



Which combination correctly identifies the type of eukaryotic DNA?

	V	W	X	Y	Z
A	centromere	intron	telomere	enhancer	5' UTR
B	centromere	enhancer	telomere	intron	5' UTR
C	intron	enhancer	telomere	5' UTR	centromere
D	telomere	enhancer	intron	5' UTR	centromere

- 20** In a series of plant breeding experiments, a pure-breeding plant with big and hairy leaves was crossed with a pure-breeding plant with small and hair-less leaves. The leaves in the F_1 generation were all big and hairy. Self-fertilisation of the F_1 generation produced the following results:

905	big and hairy leaves
301	big and hair-less leaves
305	small and hairy leaves
98	small and hair-less leaves

An F_2 plant with big and hairy leaves was crossed with an F_2 plant with small and hairy leaves. What is the maximum proportion of plants with small and hair-less leaves that could have appeared in the resulting progeny?

- A** 0%
- B** 12.5%
- C** 25%
- D** 50%

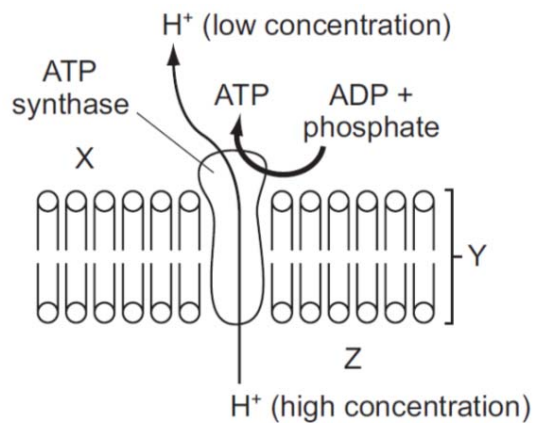
- 21** The table shows the results of a study made on a large number of twins.

Twin group	Mean difference in eye colour intensity/a.u.	Mean difference in weight/kg
Identical, raised together	1.7	2.0
Identical, raised apart.	1.8	4.8
Non-identical, same-sex, raised together	4.4	4.9

What do these results suggest about the influence of genes and environment on eye colour intensity and weight in humans?

- A** Genes have a greater influence than the environment on the eye colour intensity and the weight of identical twins.
- B** Eye colour intensity and weight are influenced by the environment.
- C** Weight is influenced by environment and genes; eye colour intensity is mainly influenced by genes.
- D** The environment has more effect than genes on the eye colour intensity and weight of non-identical twins.

22 The diagram shows a membrane in a cell.



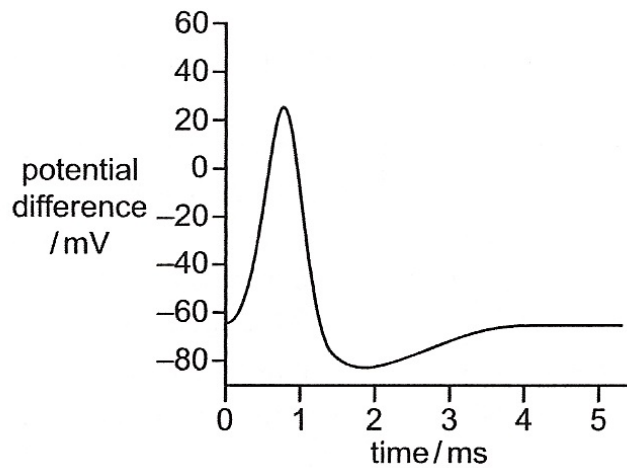
Which would be true of the diagram?

- A X is the thylakoid space, Y is the thylakoid membrane and the diagram shows ATP synthesis in a chloroplast.
 - B X is the stroma, Y is the thylakoid membrane and the diagram shows ATP synthesis in a mitochondrion.
 - C Y is the thylakoid membrane, Z is the cytosol and the diagram shows ATP synthesis in a chloroplast.
 - D Z is the intermembrane space, X is the matrix and the diagram shows ATP synthesis in a mitochondrion.
- 23 During substrate-level phosphorylation, ATP is synthesised from ADP and phosphate.

What is the immediate source of energy for this reaction?

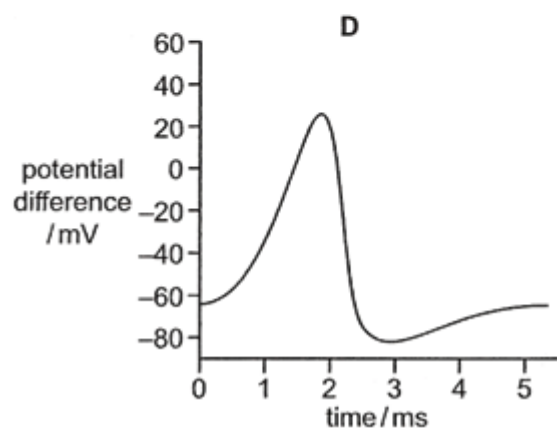
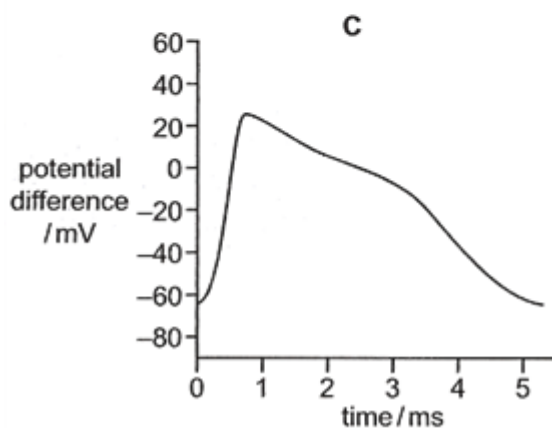
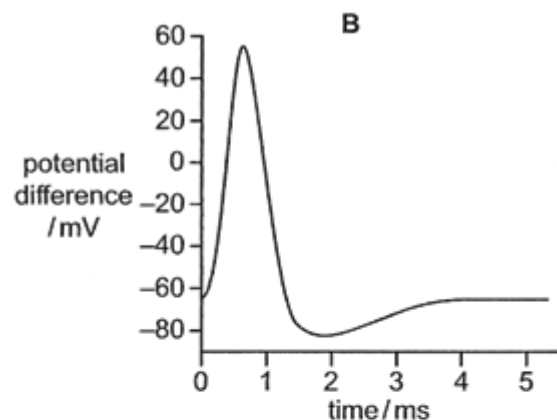
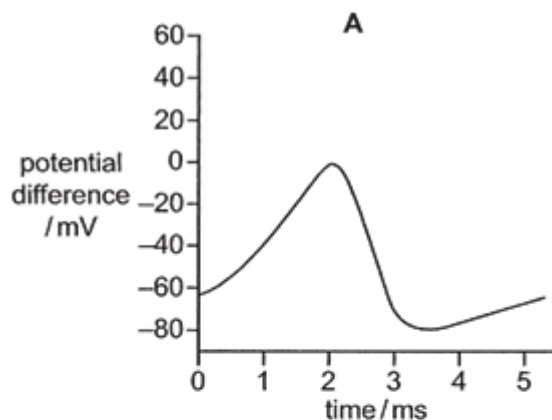
- A chemical bond energy released during the light-independent stage of photosynthesis
- B chemical bond energy released during glycolysis and the Krebs cycle
- C kinetic energy of protons diffusing through mitochondrial membranes into the mitochondrial matrix
- D kinetic energy of protons diffusing through thylakoid membranes in chloroplasts

24 The diagram shows a normal action potential.



Tetraethylammonium ions bind with and block voltage-gated K^+ channels. If these ions are present in small quantities, some of these channels in the cell surface membrane of a nerve axon are inactivated.

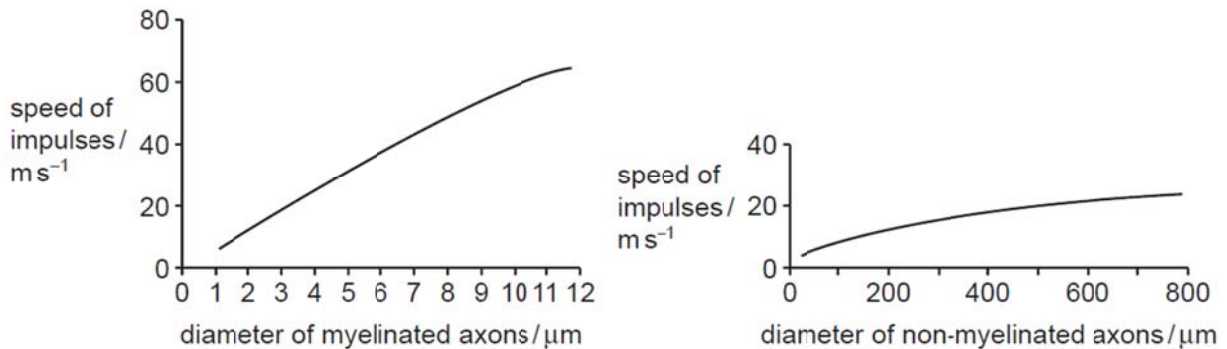
Which diagram shows an action potential in a neuron affected by a small quantity of tetraethylammonium ions?



- 25** The ability of organisms to respond rapidly to stimuli is limited by the speed of the impulses in their neurones.

The axons of invertebrate neurones lack a myelin sheath. The axons of most vertebrate neurones are myelinated.

The graphs show the speed of impulses in these two types of axon.



Which statements about these data are correct?

- 1 The action potential in myelinated axons is greater than the action potential in nonmyelinated axons.
- 2 The speed of impulses is changed by the diameter of the axon.
- 3 Increasing the diameter of a myelinated axon causes a greater change to the conduction speed than increasing the diameter of a non-myelinated axon.
- 4 The presence of myelin increases the speed at which impulses are conducted.

- A** 1, 2 and 3 only
B 1, 2 and 4 only
C 2, 3 and 4 only
D 3 and 4 only

- 26** What is necessary for an organism to respond to changes in its internal environment?

- A** A communication system between receptors and effectors.
B A communication system involving nerve cells.
C Negative feedback from receptor to effector.
D Positive feedback from effector to receptor.

27 Which of the following treatment/s could a diabetic patient undergo?

- 1 Careful control of diet
- 2 Administer insulin orally before meals
- 3 *In vitro* insertion of normal insulin gene into liver cells

- A 1 only
B 1 and 3 only
C 2 and 3 only
D 1, 2, and 3

28 Which of the following process leads to signal amplification in the cell?

- 1 Binding of multiple signal molecules to the membrane bound receptor.
- 2 Phosphorylation of the first kinase of a cascade.
- 3 Branching of different signalling pathways that produce multiple cellular responses.
- 4 Activation of numerous intracellular receptors that acts as transcription factors in gene expression.

- A 2 only
B 1 and 4 only
C 2 and 3 only
D 1, 2, 3 and 4

29 Before the settlement of California in the 1800s, the elk population was very large. By about 1900 there were only a few dozen elk left.

Owing to protection, there are now about 3000 elk living in a small number of isolated herds.

Unfortunately, some of the elk in all the herds have difficulty grazing due to a shortened lower jaw.

Which statements best explain this?

- 1 The early settlers only hunted elk that could graze.
- 2 There is random mating within each herd.
- 3 The current elk population demonstrates a genetic drift.
- 4 There was directional selection favouring short jaws.

- A 4 only
B 1 and 3 only
C 2 and 3 only
D 2 and 4 only

- 30** The frequency of recessive alleles in population is influenced by selection pressure.

Which row shows the conditions in which recessive alleles are retained in a population?

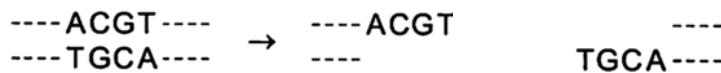
	Environmental variation of habitats	Heterozygous advantage	No selective advantage	Polymorphism (many phenotypes)
A	✓	✓	✓	✓
B	✓	x	x	✓
C	x	✓	x	x
D	x	x	✓	x

- 31** Many types of evidence, including homology, can provide evidence in support of Darwin's theory of natural selection.

What statement provides support?

- 1 The allele for sickle cell haemoglobin that gives resistance to malaria is more frequent in malarial areas.
 - 2 The distribution of the variants of the A blood group antigen reflects human migration patterns.
 - 3 The homozygous condition of the sex-linked allele for a non-functional blood clotting protein is rare.
 - 4 The molecular structure of ATP is almost identical in all eukaryotes.
- A** 4 only
B 2 and 3
C 1, 3 and 4
D 1, 2, 3 and 4

- 32 The restriction enzyme *Tai*1, cuts DNA as shown.



Which of the plasmids below can be cut using restriction enzyme *Tai*1, in order to allow DNA to be inserted into the circular plasmid without any loss of any part of the plasmid?

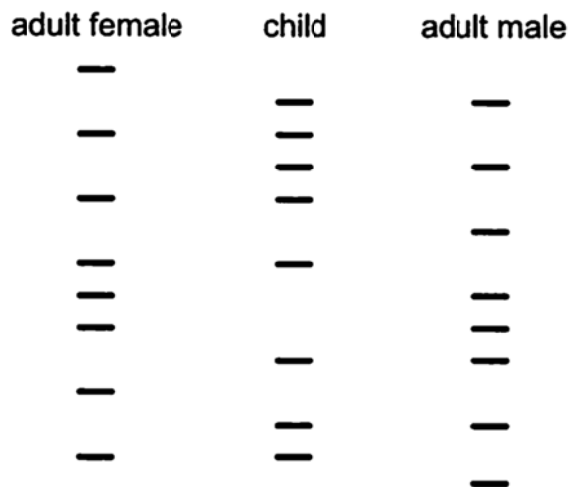
- A ----TAACGTAC----CTCAAGCT----
 ----AATGCATG----GAGTTCGA----
- B ----TTAACGTA----CACGTGGT----
 ----AATTGCAT----GTGCACCA----
- C ----ACGTACGT----TCCACGTA----
 ----TGCATGCA----AGGTGCAT----
- D ----ACCTAGGT----CCACCTGA----
 ----TGGATCCA----GGTGGACT----

key

---- other nucleotides
 The rest of the plasmid
 does not contain this
 restriction site.

- 33 In isolating the gene that produces human insulin, reverse transcriptase was used. Which is the reason that reverse transcriptase is not used for isolating the gene that produces a human sodium ion/calcium ion channel protein?
- A The amino acid sequence for this protein is not known so this is a gene that has not yet been located by the human genome project.
- B Reverse transcriptase is a viral enzyme, and in the current environment of suspicion of viruses, it is not possible to use such enzymes.
- C The antibiotic resistance genes transferred by reverse transcriptase with the insulin gene may be transferred to pathogens.
- D The ion channel protein gene is expressed at low levels in human cells so the mRNA produced is swamped by other mRNA.
- 34 Which of the following primers would allow for the copying of the single-stranded DNA sequence, 5'-ATGCCTAGGTC-3' in the polymerase chain reaction (PCR)?
- A 5' CGGAT 3'
- B 5' TACGG 3'
- C 5' GGCAU 3'
- D 5' GACCT 3'

- 35 The diagram shows the results of a DNA analysis, carried out by electrophoresis. The DNA of three individuals was profiled.



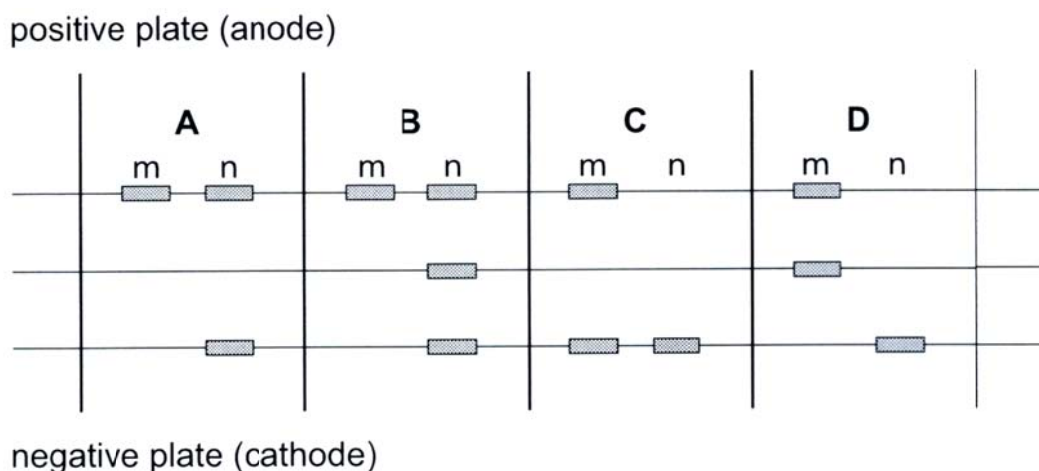
What is the most likely conclusion from these results?

- A The child cannot be related to either the male or female.
 B The female cannot be the child's mother.
 C The male cannot be the child's father.
 D The male and female could be the child's parents.
- 36 There are two forms of the seeds of the garden pea, *Pisum sativum*, one with a wrinkled skin and the other with a smooth skin.

Plants with wrinkled seeds have a recessive mutation caused by the insertion of a transposable element into a gene.

Two heterozygous pea plants were crossed. Two of the offspring (m and n) had the DNA for this gene removed and separated by electrophoresis.

Which pattern of bands shows a wrinkled phenotype and a smooth phenotype?



37 Cells produced by mitosis have several functions.

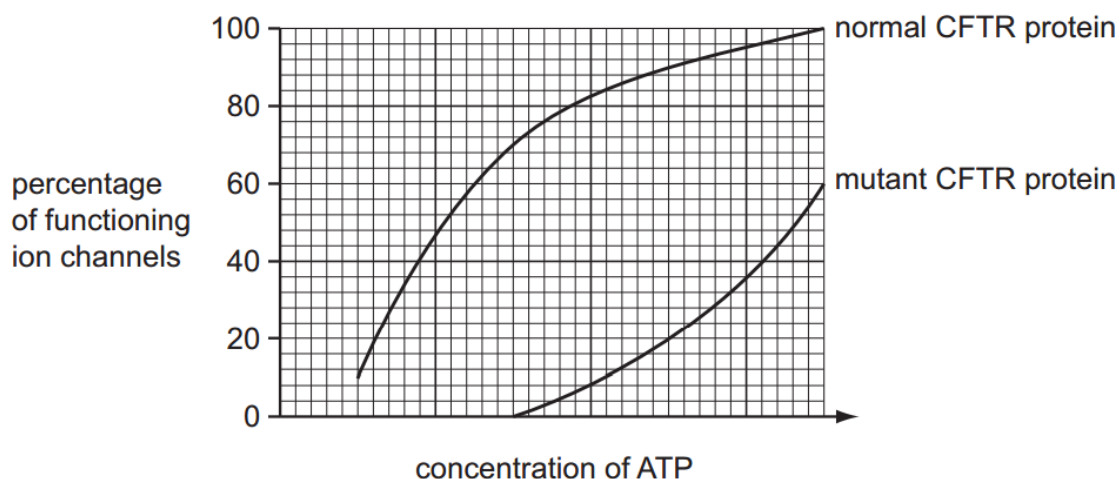
- 1 specialized cells for growth and repair
- 2 stem cells
- 3 spores produced for asexual reproduction

The genes of these cells are switched on or off during depending on their function.

Which cells have the genes promoting cell division switched on?

- A** 1 and 3 only
- B** 2 and 3 only
- C** 2 only
- D** 3 only

38 One of the many recessive mutations of the CFTR gene changes one amino acid in the region of the CFTR protein that binds ATP. The graph shows the effect of different concentrations of ATP on normal and mutant CFTR proteins.



Which correctly describes individuals who are homozygous for this mutation?

- 1 Their CFTR protein cannot bind ATP and cannot act as an ion channel.
- 2 Their CFTR protein binds ATP less readily than normal CFTR protein.
- 3 They produce CFTR protein that must bind ATP to function as an ion channel.
- 4 They produce a mixture of normal and mutant CFTR protein, both of which can act as an ion channel.

- A** 1 only
- B** 2 only
- C** 2 and 3 only
- D** 2 and 4 only

- 39 Some plant tissue culture techniques involve the production of protoplasts using leaf tissue. The tissue is treated to obtain isolated, structurally altered cells. The treatment involves incubation in a solution that contains enzymes such as cellulases and pectinases.

Which statement about protoplasts is **not** correct?

- A Protoplasts are more susceptible to microbial contamination.
 - B Protoplasts need to be maintained in a solution of the same water potential to prevent lysis.
 - C Protoplasts become pluripotent and regenerate into whole plants when provided with correct growth factors.
 - D Protoplasts are used for the production of genetically engineered plants as they take up naked DNA easily.
- 40 In some countries where rice is eaten, the lack of β -carotene in the diet can lead to blindness.

Golden rice has had two genes added to it, to make it produce β -carotene. Seed has been available since 2004, but people are cautious about the growth of genetically modified crops, fearing that these genes could be transferred to other plants.

Which of these possible reasons is the **best** argument for allowing cultivation of golden rice?

- A Many plants already produce carotene, so gene transfer will not be a problem.
- B Cultivation will be strictly controlled, so no gene transfer will occur.
- C Cultivation will occur in countries geographically remote from protesters against modification.
- D Cultivation will occur in developing countries where people are unlikely to object.